DEVOPS WINTER ARC

DAY 22 / 90

AWS Cloud — First Look & Billing Safety

Level	Beginner-friendly	**Time**	1.5–2 hours
Focus	AWS overview · account safety · billing	**Cost target**	$0 planned spend

Today's mission

We're starting AWS today, so this is an orientation day—not a heavy cloud build. Learn what AWS is, create or check an account safely, set billing alerts, and explore the console without launching chargeable infrastructure.

Scope rule: Do not repeat previous Linux, Docker, or Kubernetes assignments. Today covers only AWS introduction, account setup, billing safety, and a short console tour.

> **Important:** A $0 budget or billing alert is NOT a hard spending limit. AWS can still charge for resources you create, and alerts may be delayed. Do not launch EC2 instances, NAT Gateways, load balancers, RDS databases, or other resources today.

Part 1 — AWS in simple words (20 minutes)

• **AWS:** cloud platform offering compute, storage, databases, networking and other services.

• **Region:** a geographic area where AWS operates infrastructure.

• **Availability Zone (AZ):** an isolated location within a Region; using multiple AZs can improve availability.

• **IAM:** Identity and Access Management—controls who can access AWS and what they can do.

• **EC2:** virtual servers in the cloud.

• **S3:** object storage for files and data.

• **VPC:** your logically isolated network in AWS.

• **CloudWatch:** metrics, logs and alarms.

• **Billing & Cost Management:** pages to review usage, charges, budgets and cost trends.

Mini task: Explain AWS in your own words in 2–3 sentences. Then explain the difference between EC2 and S3.

Part 2 — Create or check your AWS account safely (25–35 minutes)

■ Open the official AWS Free Tier page and read the current terms before signing up.

■ Use your own email and accurate details. AWS may ask for payment-card and identity/phone verification. A free account does not mean every service is free.

■ Use a strong, unique password. Enable multi-factor authentication (MFA) for the root user if available.

■ Do not create access keys for the root user. Learn IAM roles/users and least privilege before future CLI work.

■ Sign in and locate the account menu, Region selector, console search and help/support area.

■ If you already have an AWS account, do not create a duplicate just for this challenge; review its security and billing settings instead.

Part 3 — Billing safety setup (20–30 minutes)

Complete these before launching any AWS resources. Console labels can change, so use the console search if needed.

■ Open **Billing and Cost Management** and review current-month charges/usage. A zero total now does not guarantee future usage will be free.

■ Open **AWS Budgets** and create a monthly cost budget for **$0.00** if permitted, or the smallest available amount. Add actual-cost and forecasted-cost alert thresholds above $0 if available.

■ Add an email address you actively check and confirm the alert recipient.

■ If available, enable billing alerts / a CloudWatch billing alarm and review the Free Tier usage page.

■ Remember: budget alerts notify you; they do not automatically stop resources or guarantee a $0 bill. Billing data and alerts may be delayed.

■ Never launch resources just to test alerts. For future labs, check pricing first and delete resources after the exercise.

Billing safety checklist

■ I know where to find current charges and usage.

■ I created or reviewed a budget and alert recipients.

■ I understand that a $0 budget is not a hard cap.

■ I will not launch services unless a future assignment explicitly requires them and I have checked costs.

Part 4 — Console exploration (20 minutes)

Open the service pages only. Do not press Create, Launch, Deploy, or any button that provisions resources.

Area	What to find	Do not do today
IAM	Users/roles, policies and MFA/security recommendations	Do not create broad permissions or root access keys.
EC2	Instance dashboard, instance types and pricing links	Do not launch an instance.
S3	What buckets and objects are; access/pricing notes	Do not create a bucket or upload sensitive data.
VPC	VPCs, subnets, route tables and security groups	Do not create a NAT Gateway or networking resources.
Billing	Bills, budgets and Free Tier usage	Do not assume a blank chart means future usage is free.

Part 5 — Beginner knowledge check (10 minutes)

1. What is cloud computing?

Suggested answer: Using computing resources over the internet instead of owning all physical hardware.

2. What is an AWS Region?

Suggested answer: A geographic area containing AWS infrastructure.

3. EC2 vs S3?

Suggested answer: EC2 provides virtual compute servers; S3 stores objects such as files and backups.

4. What does IAM do?

Suggested answer: Controls identities and permissions for AWS resources.

5. Is every AWS service free with a Free Tier account?

Suggested answer: No. Eligibility, quotas, time limits and service-specific pricing apply; check current terms.

6. Does a $0 budget prevent charges?

Suggested answer: No. It sends alerts; it is not a hard spending cap and may not be immediate.

7. Why enable MFA?

Suggested answer: It adds a verification step and makes account takeover harder.

8. What should you do after a future lab?

Suggested answer: Review resources in relevant Regions, delete what the task says to remove, and check billing/usage.

Proof of work

No GitHub project is required today. Capture only these important screenshots, hiding your email, account ID, payment details and other personal information:

• AWS Console home page (personal details hidden).

• Budget configuration showing amount and alert thresholds (hide email/account identifiers).

• Billing/Free Tier usage page, if available.

Final completion checklist

■ I can explain AWS, Region, AZ, IAM, EC2, S3, VPC and CloudWatch at a basic level.

■ I created or checked my account and enabled MFA if available.

■ I reviewed billing, created/reviewed alerts, and understand their limitations.

■ I explored the console without launching chargeable resources.

Day 22 complete: AWS account awareness + billing safety + first console tour. Today's target is safe setup and understanding—not deploying infrastructure.